POWER: USB 5V (D101) / LiPo (Q101) auto power path -> VSYS -> U6 LP5907 3.0 V -> +3V0_A (ADS1298 AVDD)
U6 EN = XIAO 3V3, so the AFE is off whenever the ESP32 is off. DVDD = XIAO 3V3.

XIAO ESP32-S3
U2
ADS_RESET 1
ADS_START 2
ADS_CS 3
ADS_DRDY 5
UART_TX 7
D0_IO1
D1_IO2
D2_IO3
D3_IO4
D4_IO5
D5_IO6
TX_IO43
BAT+
BAT-
VBAT 15
AGND 16
5V 14 VBUS
GND 13 AGND
3V3 12 +3V3
D10_IO9 11 SPI_MOSI
D9_IO8 10 SPI_MISO
D8_IO7 9 SPI_SCK
RX_IO44 8 UART_RX

B5819W
D101
VBUS
VSYS
#FLG07
PWR_FLAG

AO3401A
Q101
R104 100K
VBUS
VBAT
VSYS
C107 10uF
AGND

LP5907MFX-3.0
U6
VSYS 1
+3V3 3
IN
EN
OUT 5 +3V0_A
GND
AGND 2
C101 1uF
C102 100nF
+3V0_A
AGND

TEST POINTS
+3V0_A TP1
+3V3 TP2
AGND TP3
VSYS TP4
RLD_OUT TP5
ADS_DRDY TP6
SPI_SCK TP7
AGND TP8

LiPo (protected)
J3
1 VBAT
2 AGND
#FLG04
PWR_FLAG
VBAT
#FLG02
PWR_FLAG
AGND

LiPo: J3 wires to the XIAO underside BAT+/BAT- pads (U2 pins 15/16). XIAO is hand-soldered.

C120 100nF
C119 100nF
C118 10uF
+3V0_A
AGND

ADS1298IPAGR
U1
C121 1uF
C122 100nF
+3V3
AGND

RLD: 1M || 1.5nF feedback, 330k series to the REF electrode(s) limits DC fault current to <10 uA.
R101 1M
C111 1.5nF
R102 330K
RLD_OUT
RLD_INV
RLD_ELEC

DEBUG: USB serial over the XIAO USB-C is the default.
U5/J2 isolated UART is optional and DNP. Populate U5, C701, C702, J2 if needed;
J2 side is powered by the USB-UART adapter (3.3 or 5 V).

ISO7721DWVR (DNP)
U5
+3V3 1
UART_RX 2
UART_TX 3
AGND 4
VCC1
OUTA
INB
GND1
VCC2 8 VCC_ISO
INA 7 ISO_RX
OUTB 6 ISO_TX
GND2 5 GND_ISO

ISO UART (DNP)
J2
ISO_RX 1
ISO_TX 3
GND_ISO 2
VCC_ISO 4

C701 100nF (DNP)
+3V3
AGND
C702 100nF (DNP)
VCC_ISO
GND_ISO
#FLG05
PWR_FLAG
VCC_ISO
#FLG06
PWR_FLAG
GND_ISO

SAFETY: with electrodes on the body and USB-C plugged in, run the laptop on battery,
unplugged from mains, or use a simulated input. Charge while worn only from
an isolated (Class II) charger. Keep the isolation gap under U5 free of copper.
LAYOUT: keep U1 and its supply away from the XIAO antenna.

INPUT RC: 22k series (limits fault and plug-insertion current) + 1nF COG to AGND, ~7 kHz.
ADS1298 oversamples at 512 kHz, so this is the only anti-alias filter needed.

CH1 (TRS)
J1
S RLD_ELEC
R ELEC1_N
T ELEC1_P
ELEC1_P R201 22k CH1_P
ELEC1_N R202 22k CH1_N
C201 1nF C0G CH1_P AGND
C202 1nF C0G CH1_N AGND

CH2 (TRS)
J4
S RLD_ELEC
R ELEC2_N
T ELEC2_P
ELEC2_P R203 22k CH2_P
ELEC2_N R204 22k CH2_N
C203 1nF C0G CH2_P AGND
C204 1nF C0G CH2_N AGND

CH3 (TRS)
J5
S RLD_ELEC
R ELEC3_N
T ELEC3_P
ELEC3_P R205 22k CH3_P
ELEC3_N R206 22k CH3_N
C205 1nF C0G CH3_P AGND
C206 1nF C0G CH3_N AGND

ELECTRODES: one TRS per channel (T +, R −, S REF). All sleeves share RLD_ELEC,
so use 3 REF electrodes (9 total) or one (7 total). J6 FPC is in parallel with the jacks:
use one or the other. J6 1/5/9 + MP = AGND guards.

FLEX ARRAY (FPC 12P 0.5mm 1mm tail)
J6
AGND 1
ELEC1_P 2
ELEC1_N 3
RLD_ELEC 4
AGND 5
ELEC2_P 6
ELEC2_N 7
RLD_ELEC 8
AGND 9
ELEC3_P 10
ELEC3_N 11
RLD_ELEC 12
AGND MP

ADS1298: CH1-CH3 used, CH4-CH8 and TESTP tied to AVDD per datasheet (power down in firmware).
1, 2 or 3 channels by mode, no rework. HPF/LPF in firmware: capture ~0.5 Hz up at 4 kSPS+.
Lead-off detection and RLD channel selection are firmware settings.
Internal clock (CLKSEL high), internal 2.4 V reference.

U1
CH1_P 16
CH1_N 15
CH2_P 14
CH2_N 13
CH3_P 12
CH3_N 11
+3V0_A 10
+3V0_A 9
+3V0_A 8
+3V0_A 7
+3V0_A 6
+3V0_A 5
+3V0_A 4
+3V0_A 3
+3V0_A 2
+3V0_A 1
VCAP1 28
VCAP2 30
VCAP3 55
VCAP4 26
VREFP 24
AGND 25
IN1P
IN1N
IN2P
IN2N
IN3P
IN3N
IN4P
IN4N
IN5P
IN5N
IN6P
IN6N
IN7P
IN7N
IN8P
IN8N
VCAP1
VCAP2
VCAP3
VCAP4
VREFP
VREFN
AVDD 19 +3V0_A
AVDD1 54 +3V0_A
DVDD 48 +3V3
AVSS 20 AGND
AVSS1 53 AGND
DGND 33 AGND
WCT 64
RLDOUT 63 RLD_OUT
RLDIN 62 RLD_ELEC
RLDINV 61 RLD_INV
RLDREF 60
TESTP_PACE_OUT1 17 +3V0_A
TESTP_PACE_OUT2 18 +3V0_A
GPIO1 42
GPIO2 44
GPIO3 45
GPIO4 46
DAISY_IN 41 AGND
SCLK 40 SPI_SCK
DIN 34 SPI_MOSI
CS 39 ADS_CS
DOUT 43 SPI_MISO
CLK 37
CLKSEL 52 +3V3
START 38 ADS_START
DRDY 47 ADS_DRDY
PWDN 35 ADS_PWDN
RESET 36 ADS_RESET
RESV1 31 AGND

C112 10uF VREFP AGND
C113 100nF VREFP AGND
C114 22uF VCAP1 AGND
C115 1uF VCAP2 AGND
C116 1uF VCAP3 AGND
C117 1uF VCAP4 AGND

R103 10k +3V3 ADS_CS
R105 10k +3V3 ADS_RESET
R106 10k +3V3 ADS_PWDN

Spec: emg_headband/hardware/emg_afe/README.md
1 to 3 channel surface EMG, ADS1298 + XIAO ESP32-S3, JLCPCB assembly

Sheet: /
File: emg_wearable_revA.kicad_sch

Title: EMG wearable AFE

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